

Name: _____

PLACE VALUE AND ORDERING INTEGERS, (ESTIMATING CALCULATIONS)

Date: _____

LO: I can estimate answers to calculations by rounding numbers to one significant figure.

Estimating calculations

To estimate a calculation, round each number to one significant figure (1 s.f.) first, then perform the calculation with the rounded values.

Why estimate? Estimating checks your answer is sensible and helps spot errors.

Quick examples

●	48×31	$50 \times 30 = 1500$	round to 1 s.f.
●	$198 \div 4.1$	$200 \div 4 = 50$	round each number
●	26	$25 = 5$	round to nearest square
●	48×31	$48 \times 30 = 1440$	must round both numbers
●	3.7 rounded to 1 s.f. is 3		should be 4

Key idea

Round every number to 1 significant figure, then calculate.
The symbol $\approx$ means approximately equal to.

$$3.8 \times 19.6 \quad 4 \times 20 = 80$$

Round each value to 1 s.f. before calculating

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - multiplication estimate

Estimate 38×52

Round: 38 40, 52 50
 40×50
 $= 2000$

2000

Both numbers rounded to 1 significant figure.

Example 2 - division estimate

Estimate $412 \div 7.8$

Round: 412 400, 7.8 8
 $400 \div 8$
 $= 50$

50

Choose values that make the division easier.

Example 3 - combined operations

Estimate $(4.9 \times 21.3) \div 0.48$

Round: 4.9 5, 21.3 20, 0.48 0.5
 $(5 \times 20) \div 0.5 = 100 \div 0.5$
 $= 200$

200

Follow BIDMAS after rounding.

Example 4 - with square roots

Estimate $(63) \times 4.2$

Round: 63 $64 = 8$, 4.2 4
 8×4
 $= 32$

32

For square roots, round to the nearest perfect square.

YOUR TURN – PRACTICE (deliberate practice)

1. Rounding to 1 s.f.

Round each number to 1 significant figure.

- a) 47
- b) 683
- c) 3.62
- d) 0.078

2. Estimating multiplications

Estimate each calculation.

- a) 29×41
- b) 58×32
- c) 4.8×6.1
- d) 197×3.2

3. Estimating divisions

Estimate each division.

- a) $396 \div 8.1$
- b) $61.2 \div 2.9$
- c) $7890 \div 42$
- d) $0.82 \div 0.19$

4. Combined operations

Estimate each calculation.

- a) $(5.1 \times 9.8) \div 2.1$
- b) $(41 + 59) \times 3.8$
- c) $(312 - 97) \div 4.2$
- d) $(81.3) \times 2.9$

5. Mixed estimation

Estimate each calculation to 1 s.f.

- a) 67×4.3
- b) $891 \div 29$
- c) $(48) + 3.1^2$
- d) $(6.2)^2 \times 1.9$
- e) $5.1 \times 3.9 \div 0.21$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) 48 rounded to 1 s.f. is 50 ☐

Correct answer: _____

Reason: _____

b) 3.7 rounded to 1 s.f. is 3 ☐

Correct answer: _____

Reason: _____

c) Estimating gives an approximate answer, not exact ☐

Correct answer: _____

Reason: _____

d) 0.052 rounded to 1 s.f. is 0.1 ☐

Correct answer: _____

Reason: _____

e) 50 7 because $7^2 = 49$ ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A hall seats **48 rows** of **32 chairs**. Tickets cost **£18.50** each. Estimate the total income if the hall is full. Then calculate the exact answer. How close was your estimate?

Expression: _____

Simplified: _____

b) Estimate $62.3 \times 4.8\%$. Show your working. Then use a calculator to find the exact answer. Find the percentage error in your estimate.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

